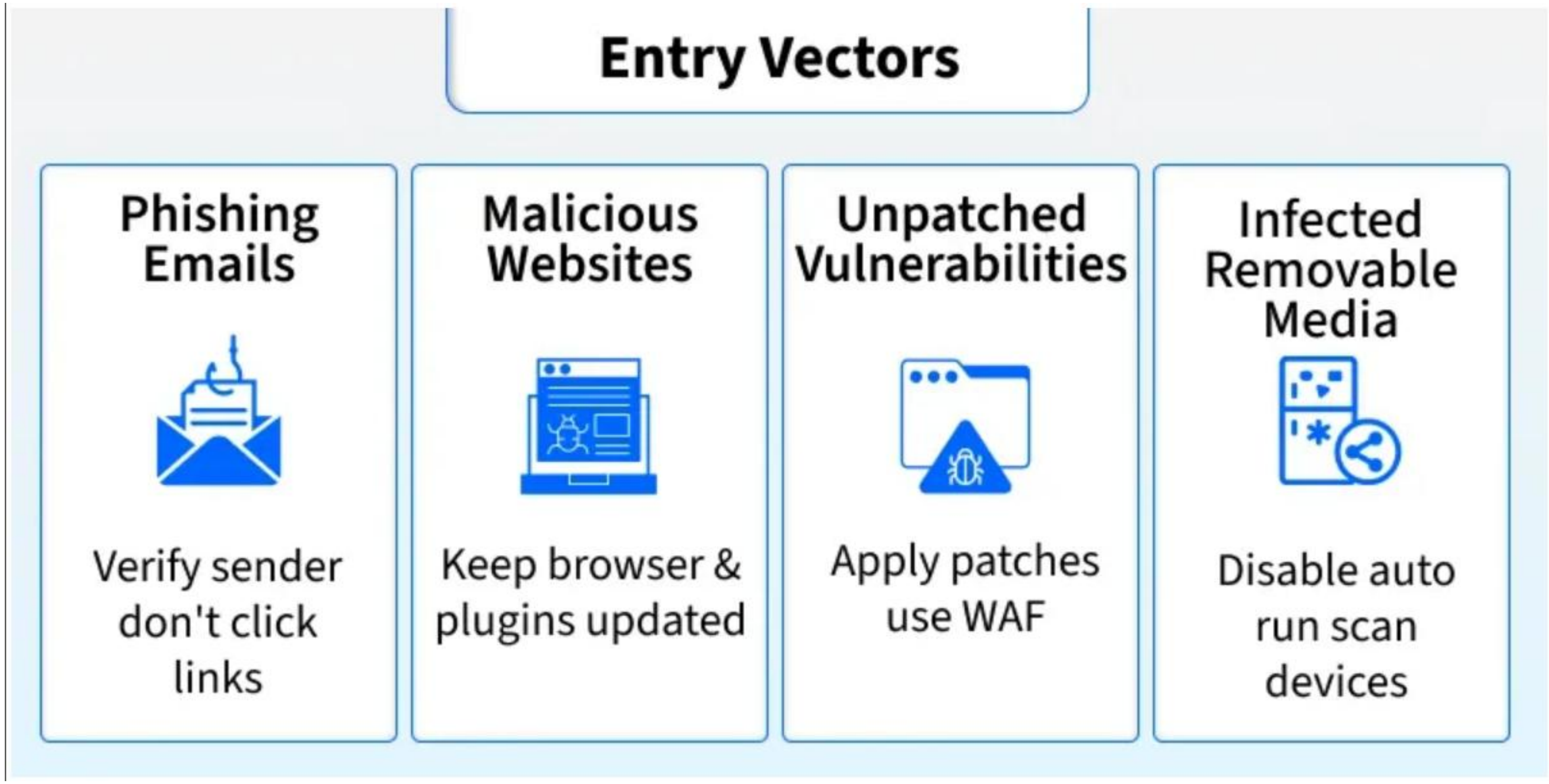
VIRUS AND MALWARE

MALWARE

- Malware is software that infects systems without user consent to steal sensitive data (bank details, passwords, personal emails), disrupt operations, or alter core system behavior.
- It can exfiltrate confidential information, corrupt or delete files, and impair system availability or integrity.
- Examples include ransomware (encrypts files for ransom) and spyware (monitors activity); mitigate risks with up-to-date antivirus and caution when opening links or attachments.

ENTRY VECTORS FOR MALWARE



TYPES OF MALWARE

- Virus: A Virus is a malicious executable code attached to another executable file. The virus spreads when an infected file is passed from system to system. Viruses can be harmless or they can modify or delete data. Opening a file can trigger a virus. Once a program virus is active, it will infect other programs on the computer.
- Worms: Worms replicate themselves on the system, attaching themselves to different files and looking for pathways between computers, such as computer network that shares common file storage areas. Worms usually slow down networks. A virus needs a host program to run but worms can run by themselves. After a worm affects a host, it is able to spread very quickly over the network.
- Trojan Horse: A Trojan horse is malware that carries out malicious operations under the appearance of a desired operation such as playing an online game. A Trojan horse varies from a virus because the Trojan binds itself to non-executable files, such as image files, and audio files.

TYPES OF MALWARES

- Ransomware: Ransomware grasps a computer system or the data it contains until the victim makes a payment. Ransomware encrypts data in the computer with a key that is unknown to the user. The user has to pay a ransom (price) to the criminals to retrieve data. Once the amount is paid the victim can resume using his/her system.
- Adware: It displays unwanted ads and pop-ups on the computer. It comes along with software downloads and packages. It generates revenue for the software distributer by displaying ads.
- Spyware: Its purpose is to steal private information from a computer system for a third party. Spyware collects information and sends it to the hacker.
- Rootkits: A rootkit modifies the OS to make a backdoor. Attackers then use the backdoor to access the computer distantly. Most rootkits take advantage of software vulnerabilities to modify system files.

TYPES OF MALWARE

- Logic Bomb: A logic bomb is a malicious program that uses a trigger to activate the malicious code. The logic bomb remains non-functioning until that trigger event happens. Once triggered, a logic bomb implements a malicious code that causes harm to a computer. Cybersecurity specialists recently discovered logic bombs that attack and destroy the hardware components in a workstation or server including the cooling fans, hard drives, and power supplies. The logic bomb overdrives these devices until they overheat or fail.
- Backdoors: A backdoor bypasses the usual authentication used to access a system. The purpose of the backdoor is to grant cyber criminals future access to the system even if the organization fixes the original vulnerability used to attack the system.
- Keyloggers: Keylogger records everything the user types on his/her computer system to obtain passwords and other sensitive information and send them to the source of the keylogging program.

SIGNS YOUR DEVICE IS INFECTED

- Performing poorly on the computer by execution.
- When your web browser directs you to a website you didn't intend to visit, this is known as a browser redirect.
- Warnings about infections are frequently accompanied by offers to buy a product to treat them.
- Having trouble starting or shutting down your computer.
- Persistent pop-up ads.

HOW TO REMOVE MALWARE

- Install Malwarebytes on the target device (Windows, macOS, Android, or iOS).
- Open the application and update its malware definitions if prompted.
- Start a scan (manual) to inspect the system — Malwarebytes scans running processes, registry entries, hard drives, and individual files.
- Review scan results when the scan completes; identified items will be listed with threat names and locations.
- Quarantine detected items to isolate suspicious files and prevent further harm.
- Remove or clean quarantined items after review to eliminate the malware from the system.
- Reboot if required to complete cleanup (follow tool prompts).
- Verify system health by re-scanning or checking for residual symptoms (performance, unexpected pop-ups).

VIRUS

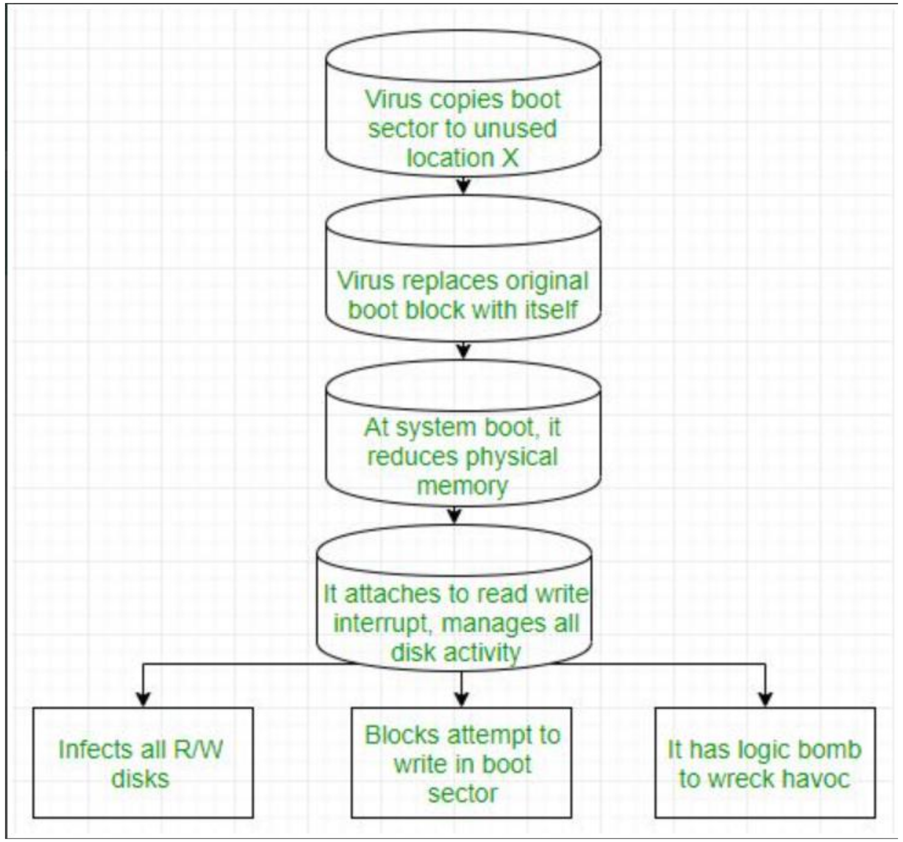
- A virus is a small piece of code hidden inside a normal program. It can make copies of itself and infect other programs.
- Viruses can damage the system by changing or deleting files, which may cause the computer to crash or not work properly.
- A virus dropper, usually a trojan horse, helps put the virus into the computer.
- In essence, a computer virus alters how your system functions and tries to infect other computers by attaching itself to legitimate software or documents that support executable code, such as macros.

TYPES OF VIRUS

- File Virus: A **File Virus** is a type of computer virus that attaches itself to executable files (such as .exe or .com files). When the infected file is run, the virus becomes active and can spread by infecting other executable files on the system. This type of virus infects the system by appending itself to the end of a file. It changes the start of a program so that the control jumps to its code.
- After executing, it usually transfers control back to the original program, making the infection less noticeable. File viruses are also known as **parasitic viruses** because they depend on host files to function but keep the host operational.

TYPES OF VIRUS

- Boot Sector Virus: A **Boot Sector Virus** is a type of virus that infects the part of a storage device that helps start the computer, called the boot sector or master boot record (MBR). When the computer turns on, this virus loads into memory before the operating system starts. Because it runs so early, it can take control of the system and is hard for antivirus programs to find and remove. It can spread through infected devices like hard drives or USB drives.
- They are sometimes referred to as **memory resident viruses** because they stay active in the system's memory after execution.



TYPES OF VIRUS

- Macro Virus: A Macro Virus is a kind of virus that is written using simple programming languages like Visual Basic. It hides inside files like Word documents or Excel spreadsheets. When you open an infected file, the virus runs automatically and can spread to other files on your computer. It may also send itself to other people through email. Because it uses common programs like Microsoft Word and Excel, it can spread quickly through shared files and attachments.
- Source Code Virus: A **Source Code Virus** is a type of virus that searches for source code files on a computer and adds harmful code to them. When the changed source code is compiled or run, the virus becomes active and can spread further. These viruses usually affect software development tools and are hard to find because they look like normal code. Their main purpose is to secretly add dangerous commands to programs while they are being created.

TYPES OF VIRUS

- Polymorphic Virus: The word "poly" means many, and "morphic" means forms. So, a Polymorphic Virus is a complicated computer virus that changes its form every time it spreads to avoid being caught by antivirus programs. It is a self encrypting virus that uses a special part called a mutation engine to constantly change its code while still doing the same harmful actions. This makes it hard for antivirus software to detect and stop it.
- Encrypted Virus: This type of virus hides by keeping its code scrambled so antivirus programs cannot easily find it. It also has a small program that unscrambles (decrypts) the code first, then the virus runs.
- Email Virus: An email virus is defined as an email that consists hidden malicious program that affects the system. The malicious program inside the email gets activated once the user opens the malicious attachments with emails or when clicking on infected links.

TYPES OF VIRUS

- Stealth Virus: A Stealth Virus is a sneaky virus that hides from antivirus programs. It changes or hides the parts of the computer that antivirus looks at. For example, when you open a file, the virus shows the clean, original file instead of the infected one. This makes it hard to find.
- Tunneling Virus: A Tunneling Virus is a virus that hides from antivirus programs by attaching itself to important parts of the computer, like system controls or drivers. This helps the virus block antivirus from finding it and keeps it active on the computer.
- Multipartite Virus: A Multipartite Virus is a virus that can infect different parts of a computer, such as files, the boot sector, and memory. Because it attacks in many ways at the same time, it is harder to detect and remove.

TYPES OF VIRUS

- Armored Virus: An Armored Virus is a virus designed to protect itself from being discovered and studied by antivirus programs. It uses tricks like hiding where it really is or making its code confusing, so it's harder for security tools to find and remove it.
- Browser Hijacker: Browser Hijacker virus is coded to target the user's browser and can alter the browser settings. It is also called the browser redirect virus because it redirects your browser to other malicious sites that can harm your computer system.
- Memory Resident Virus: A Memory Resident Virus is a virus that stays active in a computer's memory (RAM) after it runs. Because it remains in memory, it can keep running in the background, interfering with normal system operations and making it harder to detect and remove.

TYPES OF VIRUS

- Direct Action Virus: A Direct Action Virus is a virus that activates and spreads only when it is executed. Once triggered, it quickly infects files in specific locations, often targeting files in certain folders or directories. Its main goal is to replicate and cause damage during its active phase.
- Overwrite Virus: An Overwrite Virus is a type of virus that destroys the data in the files it infects by overwriting them. This usually makes the infected files partly or completely useless because the original information is lost and cannot be recovered.
- Directory Virus: A Directory Virus is a virus that changes the computer's directory information. It modifies the file paths that tell where files are stored, making it difficult to find or open those files. This type of virus is also known as a *File System Virus* or *Cluster Virus*.

TYPES OF VIRUS

- Companion Virus: A Companion Virus is a virus that creates a new file with a similar name to an existing program but with a different file extension. For example, if there is a file named Hello.exe, the virus might create a file called Hello.com. When the user runs the program, the virus runs instead, allowing it to spread and cause damage.
- FAT Virus: A **FAT Virus** targets the File Allocation Table (FAT), which is a part of the disk that keeps track of where files are stored. This virus can damage or change the FAT, making it hard to find or open files and possibly causing data loss. Because the FAT is important for file management, any damage to it can affect the whole system.

SIGNS OF VIRUS

- When unusual activity happens in your computer system.
- When you see your accounts are locked.
- Lots of emails are sent by third parties or hackers.
- The performance of the computer will be slow.
- The software automatically downloaded that infects your computer system.